Exp no:7. Pandas DataFrame Operations

Aim: To analyze customer orders using Pandas.

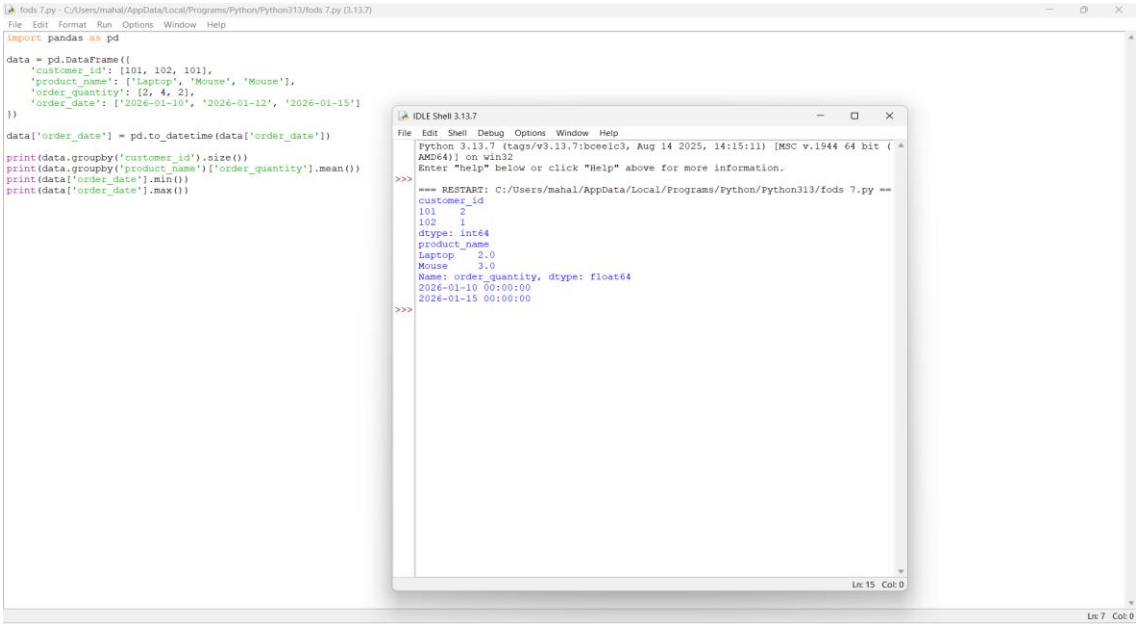
Algorithm:

1. Import Pandas.
2. Group orders by customer and product.
3. Calculate order count, average quantity, and date range.

Code:

```
import pandas as pd
data = pd.DataFrame({
    'customer_id': [101, 102, 101],
    'product_name': ['Laptop', 'Mouse', 'Mouse'],
    'order_quantity': [2, 4, 2],
    'order_date': ['2026-01-10', '2026-01-12', '2026-01-15']
})
data['order_date'] = pd.to_datetime(data['order_date'])
print(data.groupby('customer_id').size())
print(data.groupby('product_name')['order_quantity'].mean())
print(data['order_date'].min())
print(data['order_date'].max())
```

Output:



Result:

Customer orders and product quantities were analyzed successfully.